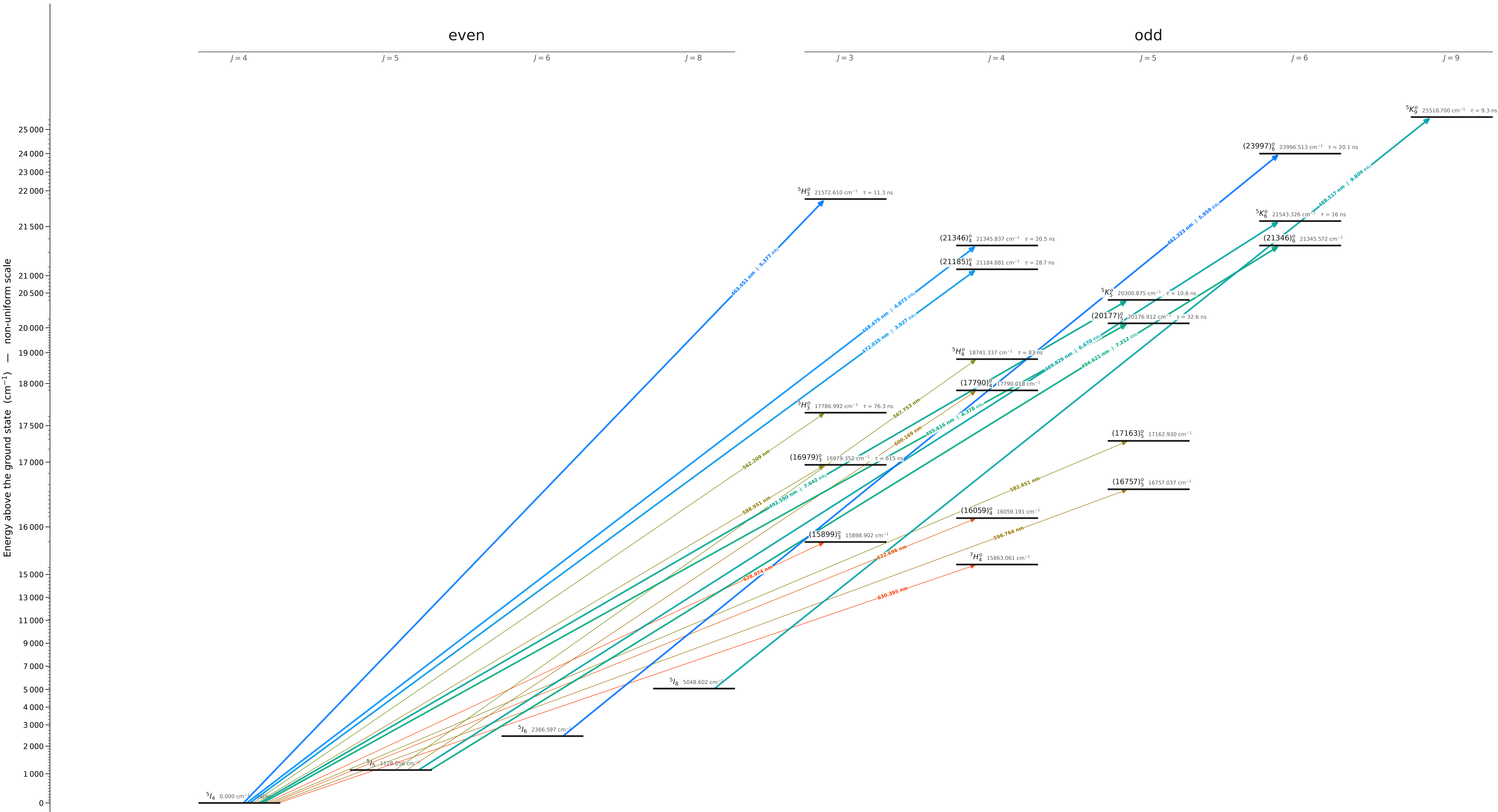
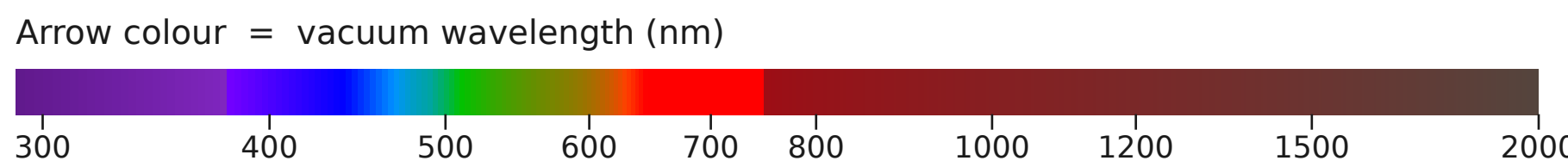


Nd⁺ ionisation limit 44560.11 cm⁻¹ = 5.52475 eV (224.416 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle U || er || J' \rangle$ in $e a_0$
Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

Measured A, B constants; sources in the data file.

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$^5\text{I}_a - ^5\text{H}_3^0$	562.209	142-150	-3048.00
$^5\text{I}_a - ^5\text{H}_2^0$	562.209	144-150	-2448.00
$^5\text{I}_a - ^5\text{H}_1^0$	562.209	146-150	-1833.00
$^5\text{I}_a - ^5\text{H}_0^0$	562.209	148-150	-1131.00
$^5\text{I}_a - ^5\text{H}_{-1}^0$	562.209	143-150	-3060.00
$^5\text{I}_a - ^5\text{H}_{-2}^0$	562.209	145-150	-2300.00
$^5\text{I}_5 - ^5\text{H}_4^0$	567.753	144-142	657.5(6)
$^5\text{I}_5 - ^5\text{H}_3^0$	567.753	146-144	586.9(5)
$^5\text{I}_5 - ^5\text{H}_2^0$	567.753	148-146	720.7(6)
$^5\text{I}_5 - ^5\text{H}_1^0$	567.753	150-148	1169.4(8)
$^5\text{I}_5 - ^5\text{H}_0^0$	567.753	143-142	257.6(5)
$^5\text{I}_5 - ^5\text{H}_{-1}^0$	567.753	145-144	190.2(4)
$^5\text{I}_a - (17163)^0_0$	582.651	144-142	-262.9(2)
$^5\text{I}_a - (17163)^0_1$	582.651	146-144	-265.8(6)
$^5\text{I}_a - (17163)^0_2$	582.651	148-146	-228.5(5)
$^5\text{I}_a - (17163)^0_3$	582.651	150-148	-127.10(20)
$^5\text{I}_a - (17163)^0_4$	582.651	143-142	-146.7(7)
$^5\text{I}_a - (17163)^0_5$	582.651	145-144	-155.2(6)
$^5\text{I}_a - (16979)^0_3$	588.951	144-142	-223.7(3)
$^5\text{I}_a - (16979)^0_4$	588.951	146-144	-227.1(3)
$^5\text{I}_a - (16979)^0_5$	588.951	148-146	-194.4(3)
$^5\text{I}_a - (16979)^0_6$	588.951	150-148	-105.80(20)

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope}).$$